# Two Unproven Effects Do Not Average to One Proven Effect: Cold Fusion and Antigravity Hybrid Claims

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**Abstract**

A recurring subgenre of fringe physics literature proposes that low-energy nuclear reactions, popularly known as cold fusion, produce as a byproduct or side effect a local reduction in gravitational weight, sometimes framed through an appeal to exotic nuclear or lattice-confinement processes said to alter the local mass-energy density in a manner that couples to gravitation. This paper argues that combining two independently unestablished claims does not strengthen either one, that the specific mechanism proposed, a coupling between anomalous nuclear heat production and gravitational weight, has no basis in the already contested cold fusion literature itself, and that the appeal of the hybrid claim rests on borrowing the more mainstream-adjacent legitimacy of ongoing low-energy nuclear reaction research to lend credibility to a gravitational claim that research has never made. The paper includes a calculation showing that the mass equivalent of any plausible excess heat is smaller than a balance could register by many orders of magnitude. It does not adjudicate the excess heat claims themselves.

## 1 Introduction

Since the 1989 announcement by Fleischmann and Pons of apparent excess heat from electrolysis of heavy water using palladium electrodes [1], low-energy nuclear reaction research has continued as a small, largely marginalized field, with proponents pointing to occasional excess-heat measurements as evidence of an unrecognized nuclear process, and the broader physics community remaining unconvinced. The early objections were concrete. Petrasso and colleagues showed that the reported gamma-ray spectrum did not have the features that a fusion product would require [2], and Lewis and colleagues searched for the neutron and other signatures of low-temperature fusion of deuterium in palladium and found none at the levels the heat would imply [3]. The United States Department of Energy convened a panel in 1989 that did not find persuasive evidence for the discovery of a new nuclear process, and a second review in 2004 reported divided opinion on the heat measurements while finding the case for a nuclear origin unpersuasive [4, 5]. Huizenga's contemporary account of the episode is the standard history of how the field's claims were evaluated [6].

A further, smaller subset of the surrounding literature layers a gravitational claim on top of the excess-heat claim, proposing that the same unspecified process reduces the weight of the reacting apparatus or of nearby objects. I have not identified a peer-reviewed primary source in which this coupling is measured. The hybrid appears chiefly in popular and self-published material, and this paper therefore treats it as a family of arguments and not as a single paper to be answered.

## 2 Why the Excess-Heat Claim Provides No Support for a Gravitational Claim

Even granting, for the sake of argument, that some genuine anomalous heat production occurs in these experiments, an anomalous heat source is not evidence of a gravitational effect unless an independent mechanism links the two, and no such mechanism is specified in this literature beyond the observation that both are,

individually, unexplained. Two unexplained phenomena occurring in the same apparatus at the same time are not evidence that they share a cause; they are, in the absence of a specified linking mechanism, exactly as likely to be two separate errors, or one error and one non-effect, as they are to be a single underlying discovery. The hybrid claim's persuasive force comes from proximity rather than argument: an apparatus already associated with one contested claim of anomalous physics is treated as more hospitable ground for a second, unrelated one.

There is also an asymmetry in the quality of the two measurements. Calorimetry of an electrolytic cell is a difficult measurement in which small systematic errors in the heat balance can mimic a modest excess, and the original dispute concerned exactly this. Weighing an electrolytic cell is difficult for a different reason, because the cell is evolving gas, losing water, changing temperature, and sitting in a convective flow, each of which changes its apparent weight. A hybrid result requires both measurements to be right and independent. It inherits the uncertainties of each.

## 3 Mass-Energy Coupling Does Not Work the Way the Hybrid Claim Requires

Some versions of this claim invoke mass-energy equivalence directly, proposing that a reaction releasing energy correspondingly reduces the reacting material's mass, and that this mass reduction manifests as reduced weight in a manner disproportionate to the actual energy released. Mass-energy equivalence is real and well confirmed, but the arithmetic shows how little it can contribute. Suppose that an experiment produced an excess of one watt, a generous figure for the table-top cells in this literature, continuously for a full day. The energy released is $86{,}400$ joules, and dividing by $c^2$, about $9 \times 10^{16}$ square meters per square second, gives a mass equivalent of about $10^{-12}$ kilograms, one nanogram. An apparatus with a mass of a hundred grams therefore changes in mass by about one part in $10^{11}$, a quantity that is below the resolution of any balance that could carry the apparatus, let alone one that could do so in the presence of gas evolution and heating.

The converse arithmetic is equally informative. To lower the weight of a hundred-gram apparatus by one percent through a mass defect, one would have to convert a gram of matter to energy, releasing about $9 \times 10^{13}$ joules, which is roughly the energy of twenty kilotons of explosive. No reported cold fusion experiment has released energy of that order, and an apparatus that had would not be a table-top curiosity. The measured excess heats in this literature, if real, correspond to mass defects that are undetectably small, and cannot be the source of any macroscopically observed weight reduction.

## 4 The Entropic Gravity Framing

On an account of gravitation as the geometric bookkeeping of a large-scale ordered configuration's redistribution, a localized, small-scale nuclear or lattice process, however anomalous its heat output might be, does not represent a redistribution at anything like the scale required to register as measurable curvature, independent of and in addition to the mass-energy equivalence argument above. Both the standard and the entropic account agree that whatever is or is not happening inside a palladium electrode, it is not happening at a scale that could plausibly touch gravitation.

## 5 Objections

Proponents occasionally argue that the relevant coupling is not through simple mass-energy equivalence but through some exotic nuclear or lattice-confinement effect not captured by standard theory, analogous to

the appeal made by other devices surveyed in this series to unspecified subtle fields. As with those cases, an appeal to an unspecified mechanism is not an argument for a mechanism, and the burden of specifying and testing it remains with the claim, a burden the cold fusion antigravity literature has not, in over three decades, discharged.

A second objection holds that the cold fusion field has been unfairly treated, that some laboratories continue to report excess heat, and that its marginalization is sociological. This paper takes no position on the excess heat claims and does not need to. If the heat is real, the argument above about mass-energy still applies. If the heat is not, there is nothing to couple. The hybrid claim fails in either case, and so its fate does not depend on how the dispute over the underlying field is resolved.

## 6 Conclusion

Layering a gravitational claim onto an already contested excess-heat claim does not strengthen either, since no specified mechanism links the two beyond the fact that both remain unexplained by mainstream theory, and the mass-energy magnitudes involved in even the most generous reading of the underlying excess-heat literature are far too small to produce any measurable gravitational effect under either the standard or the entropic account of gravitation. This paper offers no judgment of the underlying calorimetry and no new measurement. It states that the combination of two unproven effects has no evidential force greater than the weaker of the two.

## References


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